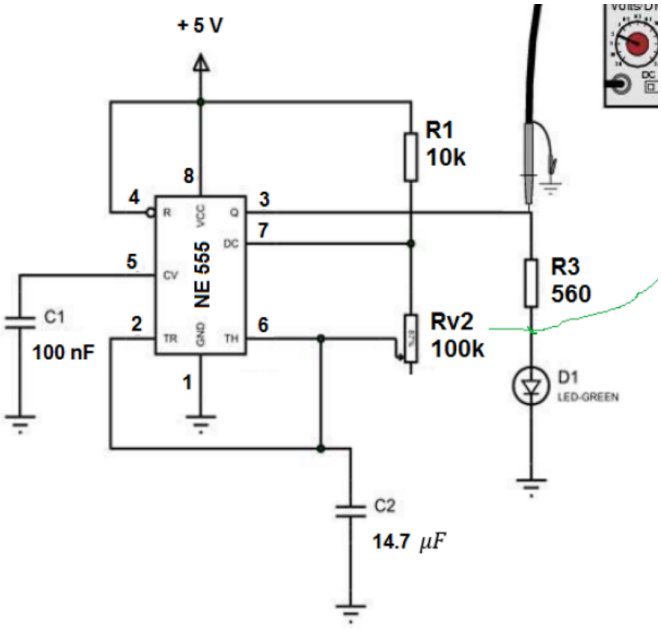


Figure 4: Oscillator-Timer schematic

In this experiment we used these components listed below.

- 1X NE555 IC
- 1X 10kΩ resistor
- 1X 560Ω resistor
- 1X 4.7μF capacitor
- 1X 10μF capacitor
- 1X 100nF capacitor
- 1X Potentiometer 100kΩ
- 1X LED
- 6X Jumper cables
- 1X Breadboard

After constructing the circuit for the second experiment we have the circuit shown in the Figure 5.

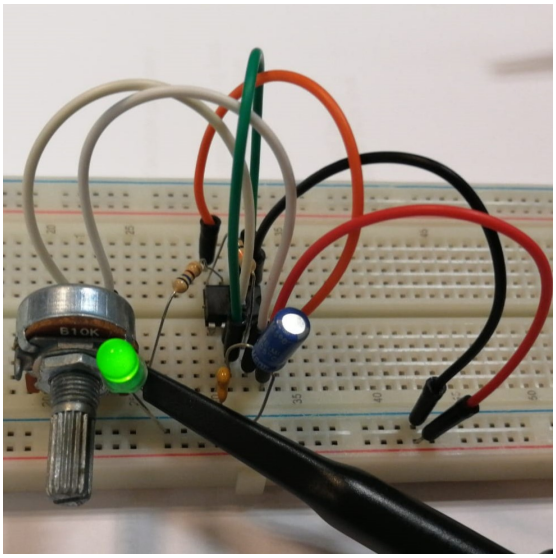


Figure 5: Oscillator-Timer schematic

We have connected oscilloscope to the one leg of the LED and captured the signal on the leg as shown in the Figure 5.



Figure 6: While potentiometer resistance is low

After measuring that signal from the circuit we have increased the resistance value of the potentiometer, and we have obtained the signals shown in the Figure 6. On the screen of oscilloscope we can see our signal frequency is 11.43Hz.